SOFTWARE ENGINEERING

LAB 7: Fuzzing for Input Validation Bugs

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Objective:

To explore fuzz testing as a security and quality assurance technique by identifying and fixing bugs in the python script provided to you using the Hypothesis fuzzing library.

SCREENSHOTS:

SS1 – Original program output:

```
PS C:\Users\khush\OneDrive\Desktop\L6_L7_56_course_registration\LAB7> py course_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs 301
Formatted course code: CS301

==== Running total_credits ====
Enter course credits separated by commas (e.g. 4,3,4): 4,3,4
Total credits: 11

==== Running parse_student_record ====
Enter a student record as SRN|name|status (e.g. pes1ug23cs001|Asha|regular): pes1ug23cs001|Asha|regular
Student record: {'srn': 'PES1UG23CS001', 'name': 'Asha', 'status': 'regular'}
PS C:\Users\khush\OneDrive\Desktop\L6_L7_56_course_registration\LAB7>
```

SS2 – Failing Hypothesis tests:

```
========================== short test summary info ==========================
FAILED test_course_registration_parser.py::test_format_course_code_no_crash
- ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_course_registration_parser.py::test_total_credits_no_crash - Exc
eptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_course_registration_parser.py::test_parse_student_record_no_cras
h - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================= 3 failed in 9.08s =============================
PS C:\Users\khush\OneDrive\Desktop\L6_L7_56_course_registration\LAB7> py -m
pytest test_course_registration_parser.py -v
```

SS3 – Fixed program output:

```
PS C:\Users\khush\OneDrive\Desktop\L6_L7_56_course_registration\LAB7> py cou
rse_registration_parser.py
==== Running format_course_code ====
Enter a course code as DEPT NUMBER (e.g. cs 301): cs 301
Formatted course code: CS301

==== Running total_credits ====
Enter course credits separated by commas (e.g. 4,3,4): 4,3,4
Total credits: 11

==== Running parse_student_record ====
Enter a student record as SRN|name|status (e.g. pes1ug23cs001|Asha|regular):
 pes1ug23cs001|Asha|regular
Student record: {'srn': 'PES1UG23CS001', 'name': 'Asha', 'status': 'regular'
}
```

SS4 – Passing Hypothesis tests:

